Problem A. tumi toh, amari

Time Limit 1000 ms
Mem Limit 262144 kB

Luke likes to eat. There are n piles of food aligned in a straight line in front of him. The i-th pile contains a_i units of food.

Luke will walk from the 1-st pile towards the n-th pile, and he wants to eat every pile of food without walking back. When Luke reaches the i-th pile, he can eat that pile if and only if $|v - a_i| \leq x$, where x is a fixed integer, and v is Luke's food affinity.

Before Luke starts to walk, he can set v to any integer. Also, for each i ($1 \leq i \leq n$), Luke can *change* his food affinity to any integer **before** he eats the i-th pile.

Find the minimum number of *changes* needed to eat every pile of food.

Note that the initial choice for v is **not** considered as a change.

Input

The input consists of multiple test cases. The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases. The description of test cases follows.

For each test case, the first line contains two integers, n, x ($1 \leq n \leq 2 \cdot 10^5$, $1 \leq x \leq 10^9$) — the number of piles, and the maximum difference between the size of a pile and Luke's food affinity, such that Luke can eat the pile.

The second line contains n integers $a_1, a_2, \ldots, a_n$ ($1 \leq a_i \leq 10^9$).

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output an integer on a separate line, which is the minimum number of changes needed.

Examples

Input	Output
7 5 3 3 8 5 6 7 5 3 3 10 9 8 7 12 8 25 3 3 17 8 6 1 16 15 25 17 23 10 2 1 2 3 4 5 6 7 8 9 10 8 2 2 4 6 8 6 4 12 14 8 2 2 7 8 9 6 13 21 28 15 5 11 4 13 23 7 10 5 21 20 11 17 5 29 16 11	0 1 2 1 2 4 6

Note

In the first test case, Luke can set v to 5 before he starts to walk. And he can walk straight to eat every piles of food without changing v.

In the second test case, Luke can set v to 3 before he starts to walk. And he could change v to 10 before he eats the second pile. After that, he can walk straight to eat remaining food without changing v.

In the fourth test case, Luke can set v to 3 before he starts to walk. And he could change v to 8 before he eats the sixth pile. After that, he can walk straight to eat remaining food without changing v.

In the fifth test case, Luke can set v to 4 before he starts to walk. And he could change v to 6 before he eats the fourth pile. Then he could change v to 12 before he eats the seventh pile. After that, he can walk straight to eat remaining food without changing v.

Problem B. chai na meye

Time Limit 1000 ms
Mem Limit 262144 kB

In Berland it is the holiday of equality. In honor of the holiday the king decided to equalize the welfare of all citizens in Berland by the expense of the state treasury.

Totally in Berland there are n citizens, the welfare of each of them is estimated as the integer in a_i burles (burle is the currency in Berland).

You are the royal treasurer, which needs to count the minimum charges of the kingdom on the king's present. The king can only give money, he hasn't a power to take away them.

Input

The first line contains the integer n ($1 \le n \le 100$) — the number of citizens in the kingdom.

The second line contains n integers $a_1, a_2, ..., a_n$, where a_i ($0 \le a_i \le 10^6$) — the welfare of the i-th citizen.

Output

In the only line print the integer S — the minimum number of burles which are had to spend.

Examples

Input	Output
5 0 1 2 3 4	10

Input	Output
5 1 1 0 1 1	1

Input	Output
3 1 3 1	4

Input	Output
1 12	0

Note

In the first example if we add to the first citizen 4 burles, to the second 3, to the third 2 and to the fourth 1, then the welfare of all citizens will equal 4.

In the second example it is enough to give one burle to the third citizen.

In the third example it is necessary to give two burles to the first and the third citizens to make the welfare of citizens equal 3.

In the fourth example it is possible to give nothing to everyone because all citizens have 12 burles.

Problem C. e ridoy, tomari...

Time Limit 1000 ms
Mem Limit 262144 kB

Codeforces separates its users into 4 divisions by their rating:

- For Division 1: $1900 \le \mathrm{rating}$
- For Division 2: $1600 \le \mathrm{rating} \le 1899$
- For Division 3: $1400 \le \mathrm{rating} \le 1599$
- For Division 4: $\mathrm{rating} \le 1399$

Given a rating, print in which division the rating belongs.

Input

The first line of the input contains an integer t ($1 \le t \le 10^4$) — the number of testcases.

The description of each test consists of one line containing one integer rating ($-5000 \le \mathrm{rating} \le 5000$).

Output

For each test case, output a single line containing the correct division in the format "`Division X`", where X is an integer between 1 and 4 representing the division for the corresponding rating.

Examples

Input	Output
7 -789 1299 1300 1399 1400 1679 2300	Division 4 Division 4 Division 4 Division 4 Division 3 Division 2 Division 1

Note

For test cases $1-4$, the corresponding ratings are -789, 1299, 1300, 1399, so all of them are in division 4.

For the fifth test case, the corresponding rating is 1400, so it is in division 3.

For the sixth test case, the corresponding rating is 1679, so it is in division 2.

For the seventh test case, the corresponding rating is 2300, so it is in division 1.

Problem D. pabe na keu tomke

Time Limit 2000 ms
Mem Limit 262144 kB

You are given an integer x. Your task is to find any integer y $(1 \le y < x)$ such that $\gcd(x, y) + y$ is maximum possible.

Note that if there is more than one y which satisfies the statement, you are allowed to find any.

$\gcd(a, b)$ is the Greatest Common Divisor of a and b. For example, $\gcd(6, 4) = 2$.

Input

The first line contains a single integer t ($1 \le t \le 1000$) — the number of test cases.

Each of the following t lines contains a single integer x ($2 \le x \le 1000$).

Output

For each test case, output any y ($1 \le y < x$), which satisfies the statement.

Examples

Input	Output
7 10 7 21 100 2 1000 6	5 6 18 98 1 750 3

Problem E. tumi onno karo hou

Time Limit 3000 ms
Mem Limit 262144 kB
Input File stdin
Output File stdout

A company has n employees numbered from 1 to n. Each employee either has no immediate manager or exactly one immediate manager, who is another employee with a different number. An employee A is said to be the superior of another employee B if at least one of the following is true:

- Employee A is the immediate manager of employee B
- Employee B has an immediate manager employee C such that employee A is the superior of employee C.

The company will not have a managerial cycle. That is, there will not exist an employee who is the superior of his/her own immediate manager.

Today the company is going to arrange a party. This involves dividing all n employees into several groups: every employee must belong to exactly one group. Furthermore, within any single group, there must not be two employees A and B such that A is the superior of B.

What is the minimum number of groups that must be formed?

Input

The first line contains integer n ($1 \le n \le 2000$) — the number of employees.

The next n lines contain the integers p_i ($1 \le p_i \le n$ or $p_i = -1$). Every p_i denotes the immediate manager for the i-th employee. If p_i is -1, that means that the i-th employee does not have an immediate manager.

It is guaranteed, that no employee will be the immediate manager of him/herself ($p_i \ne i$). Also, there will be no managerial cycles.

Output

Print a single integer denoting the minimum number of groups that will be formed in the party.

Examples

Input	Output
5 -1 1 2 1 -1	3

Note

For the first example, three groups are sufficient, for example:

- Employee 1
- Employees 2 and 4
- Employees 3 and 5

Problem F. oo..oo..oo..oo

Time Limit 1000 ms
Mem Limit 262144 kB

You are given three digits a, b, c. Two of them are equal, but the third one is different from the other two.

Find the value that occurs exactly once.

Input

The first line contains a single integer t ($1 \le t \le 270$) — the number of test cases.

The only line of each test case contains three digits a, b, c ($0 \le a, b, c \le 9$). Two of the digits are equal, but the third one is different from the other two.

Output

For each test case, output the value that occurs exactly once.

Examples

Input	Output
10 1 2 2 4 3 4 5 5 6 7 8 8 9 0 9 3 6 3 2 8 2 5 7 7 7 7 5 5 7 5	1 3 6 7 0 6 8 5 5 7

Problem G. tumi karo nou...

Time Limit 1000 ms

Mem Limit 262144 kB

Alica and Bob are playing a game.

Initially they have a binary string s consisting of only characters `0` and `1`.

Alice and Bob make alternating moves: Alice makes the first move, Bob makes the second move, Alice makes the third one, and so on. During each move, the current player must choose two **different adjacent** characters of string s and delete them. For example, if $s = 1011001$ then the following moves are possible:

1. delete s_1 and s_2: $\mathbf{10}11001 \rightarrow 11001$;
2. delete s_2 and s_3: $1\mathbf{01}1001 \rightarrow 11001$;
3. delete s_4 and s_5: $101\mathbf{10}01 \rightarrow 10101$;
4. delete s_6 and s_7: $10110\mathbf{01} \rightarrow 10110$.

If a player can't make any move, they lose. Both players play optimally. You have to determine if Alice can win.

Input

First line contains one integer t ($1 \le t \le 1000$) — the number of test cases.

Only line of each test case contains one string s ($1 \le |s| \le 100$), consisting of only characters `0` and `1`.

Output

For each test case print answer in the single line.

If Alice can win print `DA` (YES in Russian) in any register. Otherwise print `NET` (NO in Russian) in any register.

Examples

Input	Output
3 01 1111 0011	DA NET NET

Note

In the first test case after Alice's move string s become empty and Bob can not make any move.

In the second test case Alice can not make any move initially.

In the third test case after Alice's move string s turn into 01. Then, after Bob's move string s become empty and Alice can not make any move.

Problem H. tomake chara ami

Time Limit 1000 ms
Mem Limit 262144 kB

You are given two positive integers n ($1 \le n \le 10^9$) and k ($1 \le k \le 100$). Represent the number n as the sum of k positive integers of the same parity (have the same remainder when divided by 2).

In other words, find $a_1, a_2, \ldots, a_k$ such that all $a_i > 0$, $n = a_1 + a_2 + \ldots + a_k$ and either all a_i are even or all a_i are odd at the same time.

If such a representation does not exist, then report it.

Input

The first line contains an integer t ($1 \le t \le 1000$) — the number of test cases in the input. Next, t test cases are given, one per line.

Each test case is two positive integers n ($1 \le n \le 10^9$) and k ($1 \le k \le 100$).

Output

For each test case print:

- YES and the required values a_i, if the answer exists (if there are several answers, print any of them);
- NO if the answer does not exist.

The letters in the words YES and NO can be printed in any case.

Examples

Input	Output
8 10 3 100 4 8 7 97 2 8 8 3 10 5 3 1000000000 9	YES 4 2 4 YES 55 5 5 35 NO NO YES 1 1 1 1 1 1 1 1 NO YES 3 1 1 YES 111111110 111111110 111111110 111111110 111111110 111111110 111111110 111111110 111111120

Problem I. jano na... o..o..o..

Time Limit 1000 ms
Mem Limit 262144 kB

You are given two arrays a and b both of length n.

You will merge[†] these arrays forming another array c of length $2 \cdot n$. You have to find the maximum length of a subarray consisting of equal values across all arrays c that could be obtained.

[†] A merge of two arrays results in an array c composed by successively taking the first element of either array (as long as that array is nonempty) and removing it. After this step, the element is appended to the back of c. We repeat this operation as long as we can (i.e. at least one array is nonempty).

Input

Each test contains multiple test cases. The first line of input contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. The description of test cases follows.

The first line of each test case contains a single integer n ($1 \le n \le 2 \cdot 10^5$) — the length of the array a and b.

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 2 \cdot n$) — the elements of array a.

The third line of each test case contains n integers $b_1, b_2, \ldots, b_n$ ($1 \le b_i \le 2 \cdot n$) — the elements of array b.

It is guaranteed that the sum of n across all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output the maximum length of a subarray consisting of equal values across all merges.

Examples

Input	Output
4 1 2 2 3 1 2 3 4 5 6 2 1 2 2 1 5 1 2 2 2 2 2 1 1 1 1	2 1 2 5

Note

In the first test case, we can only make $c = [2, 2]$, thus the answer is 2.

In the second test case, since all values are distinct, the answer must be 1.

In the third test case, the arrays c we can make are $[1, 2, 1, 2]$, $[1, 2, 2, 1]$, $[2, 1, 1, 2]$, $[2, 1, 2, 1]$. We can see that the answer is 2 when we choose $c = [1, 2, 2, 1]$.

Problem J. bujhi na kono kichu je

Time Limit 2000 ms
Mem Limit 262144 kB

Suneet and Slavic play a card game. The rules of the game are as follows:

- Each card has an integer value between 1 and 10.
- Each player receives 2 cards which are face-down (so a player doesn't know their cards).
- The game is turn-based and consists **exactly of two turns**. In a round, both players pick a **random unflipped** card and flip it. The player who flipped a card with a strictly greater number wins the round. In case of equality, no one wins the round.
- A player wins a game if he wins the most number of rounds (i.e. strictly greater than the other player). In case of equality, no one wins the game.

Since Suneet and Slavic aren't best friends, you need to calculate the number of ways the game could happen that Suneet would end up as the winner.

For a better understanding, please check the notes section.

Input

The first line contains an integer t ($1 \le t \le 10^4$) — the number of test cases.

The first and only line of each test case contains 4 integers a_1, a_2, b_1, b_2 ($1 \le a_1, a_2, b_1, b_2 \le 10$) where a_1 and a_2 represent the cards Suneet has, and b_1 and b_2 represent the cards Slavic has, respectively.

Output

For each test case, output a single integer — the number of games Suneet would win considering all possible games.

Examples

Input	Output
5 3 8 2 6 1 1 1 1 10 10 2 2 1 1 10 10 3 8 7 2	2 0 4 0 2

Note

Consider the first test case when Slavic starts with the cards that have the values 2 and 6, and Suneet starts with cards that have the values 3 and 8. The game could happen in 4 different ways:

- Suneet flips 3 and Slavic flips 2. Suneet wins the first round. Then, Suneet flips 8 and Slavic flips 6. Suneet wins the second round as well. Since Suneet won 2 rounds, he wins the game.
- Suneet flips 3 and Slavic flips 6. Slavic wins the first round. Then, Suneet flips 8 and Slavic flips 2. Suneet wins the second round. Nobody wins since both players won an equal amount of rounds.
- Suneet flips 8 and Slavic flips 6. Suneet wins the first round. Then, Suneet flips 3 and Slavic flips 2. Suneet wins the second round as well. Since Suneet won 2 rounds, he wins the game.
- Suneet flips 8 and Slavic flips 2. Suneet wins the first round. Then, Suneet flips 3 and Slavic flips 6. Slavic wins the round. Nobody wins since both players won an equal amount of rounds.